

HOW THE USE OF AI IS CHANGING THE ROLE OF EDUCATORS AT UNIVERSITIES - AND WHY THIS IS BY NO MEANS A BAD THING

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Abstract

Higher education plays a pivotal role in shaping individuals' intellectual and professional development. In an era characterized by rapid technological advancements, economic shifts, and global challenges, it is crucial for universities to adapt their curricula to equip students with the skills necessary for the future. These skills, often referred to as "future skills," encompass a set of competencies that go beyond subject-specific knowledge and are essential for success in the evolving workforce and society. Future skills are accordingly a diverse set of competencies that enable individuals to navigate an increasingly complex world effectively. These skills-set includes the so-called digital literacy, which encompasses a range of skills, knowledge, and competencies that enable individuals to access, evaluate, create, and communicate information using digital tools (Ehlers 2020; Schüller et al. 2019;). However, digital literacy as a future skill is increasingly the subject of debate, as the rapid spread of AI is raising the question for many universities as to which core skills students actually still need and what AI can do for them in terms of academic work (Wannemacher & Bodmann 2021). No one would have thought in the current fall of 2023 that we would find ChatGPT almost a normality and that there would be a multitude of similar AI's by now. Jenny AI, Jasper or Copy AI are just a few examples of how scientific writing and scientific work are significantly influenced by AI today. For universities, this does not only pose problems with regard to authorship and author issues or plagiarism checks. On a didactic level, many university teachers feel that their role is being sensitively disturbed by AI-based writing tools. If technical questions can also be posed to the AI or technical discussions can be held with ChatGPT, who still needs the expertise of university lecturers? The paper takes up this, deliberately striking question, by critically analyzing the classical understanding of education at universities, especially at German universities. As a counter-proposal, a co-constructive educational framework is introduced, which enables educators to change their role (Weimann-Sandig 2023b). The main difference between constructivism (Vygotskij 1964) and co-constructivism lies in the emphasis on the social element, i.e. a two-way teaching-learning relationship. While constructivism focuses on individual learning, co-constructivism emphasizes the importance of social interactions and the joint construction of knowledge in social contexts. In the co-constructivist concept, lecturers also become knowledge recipients and thus continue to educate themselves through the input of their students (Weimann-Sandig 2023b). This role change offers them the possibility of a fundamental further development of their understanding of teaching and makes them adaptable to the needs of the universities of the future. In a best practice example from our university development project BediRa, we provide didactic recommendations and exercises for teachers to face the new challenge of AI in teaching and try it out together with students.

Keywords: AI, Higher Education, Higher Education Didactics, Role of Educators

1 INTRODUCTION

In recent years, AI has gained significant traction in education, with its potential to personalize learning, automate administrative tasks, and provide timely feedback to students (Crompton & Burke 2023). AI-powered educational tools and platforms have emerged, ranging from intelligent tutoring systems to virtual learning assistants, which assist students in their learning journeys (Chen et al. 2020). These advancements have sparked debates and discussions about the impact of AI on the traditional roles of educators and the future of education. AI technologies, such as machine learning algorithms and natural language processing, are increasingly being integrated into educational practices, reshaping the role of educators in the process. This paper aims to explore how the use of AI is changing the role of educators

at universities and argues that this transformation is not inherently negative but rather presents new opportunities for enhancing the learning experience. Traditionally, educators have played a central role in facilitating learning, imparting knowledge, and guiding students' academic growth. However, the integration of AI technologies has led to a shift in focus, with educators now assuming more facilitative and mentorship roles (Weimann-Sandig 2023a). By examining the evolving role of educators in the context of AI, this paper seeks to shed light on the positive aspects of this transformation. It argues that the use of AI in education can lead to improved learning outcomes and enhanced pedagogical practices.

2 DEFINITION AND OVERVIEW OF AI IN HIGHER EDUCATION

Artificial Intelligence (AI) encompasses a range of technologies and methodologies that enable machines to mimic human intelligence and perform tasks that traditionally require human cognition. In the context of education, AI refers to the application of these technologies to enhance teaching, learning, and administrative processes (Holmes 2019). AI in education encompasses various subfields, including machine learning, natural language processing, computer vision, and expert systems. Machine learning algorithms enable systems to learn from data and make predictions or decisions without being explicitly programmed (Fahd et al. 2022). Natural language processing allows machines to understand and generate human language, facilitating communication between students and AI-powered educational tools. Even if the public always talks about THE artificial intelligence, it is important to point out the differences - especially in the education sector. A distinction must be made between generative artificial intelligence and intelligent tutoring systems, for example (Nuxoll 2023). Although both technologies are based on AI, they have different areas of application and also influence the teaching and learning behavior of students and teachers differently. The best-known generative AI system is certainly ChatGPT. Its major advantage is its ability to create texts that can be adapted to human writing styles. In fact, ChatGPT in particular can be trained to copy personal writing styles. On the one hand, this makes it easier for the writer to interact with the AI, but on the other hand, of course, it also makes it increasingly difficult to distinguish which individual contribution can be attributed to the writer. This means that generative AI systems in particular affect the basic understanding of scientific work.

Intelligent Tutoring Systems (ITS) use specialized AI to provide personalized learning. As a rule, an ITS contains a model of the subject area to be taught, is based on a specific learning model and focuses on one or more types of learner (Nuxoll 2023). The great advantage of intelligent tutoring systems lies in the documentation and simultaneous analysis of students' learning progress. Not only can learning deficits be made visible, but students can also be suggested ways of compensating for these deficits. Depending on their learning type, students can choose how they can compensate for these deficits (e.g. playfully using flashcards, visualized using learning models or text-oriented by working on cloze texts). At the end, students can then use examination tasks to show the extent to which they have been able to close the gaps in their knowledge. AI in education has seen significant growth in recent years, driven by advancements in computing power, data availability, and algorithmic techniques (Weßels 2023). Other important AI applications in the field of higher education today include Automated Essay Scoring, where AI algorithms can be used to automatically evaluate and score essays. These systems analyze factors such as grammar, coherence, and content to provide objective and consistent grading. Turnitin's GradeMark and ETS's e-rater are examples of automated essay scoring tools. Some universities already use virtual learning assistants what means AI-powered chatbots or voice-activated assistants that provide support to students. These assistants can answer questions, provide explanations and offer study recommendations. VR and AR technologies enhance learning experiences by creating immersive and interactive environments and are increasingly being used in a wide variety of subjects. Even in person-centered subjects such as social work, students are now training communication and intercultural skills using VR or AR (Weimann-Sandig 2023c).

Nevertheless, the attitude of many university lecturers is divided when it comes to the achievements of AI. Looking at the debates in Germany - but also in other countries - there have been bans on the use of generative AI by students at some universities (e.g. Paris Institute of Political Science (Sciences Po)) as well as the abolition of bachelor's theses and dissertations (e.g. Faculty of Business Administration at the University of Economics in Prague). The latter is due to the realization that the use of generative AI makes it increasingly difficult to control the students' own contribution to traditional theses (Weimann-Sandig 2023a). While the integration of AI in education brings numerous benefits, it also presents certain challenges and concerns that need to be addressed for responsible implementation. Two things in particular should be emphasized at this point, as in my work with university lecturers they are repeatedly

raised as central reservations about the use of AI and must be taken absolutely seriously. At the same time, it should be pointed out that there are certainly even more reasons to take a critical look at AI in higher education. This description therefore does not claim to be exhaustive. The ethical and legal dimension of the use of AI in educational contexts should not be underestimated. As student data is collected, analyzed, and utilized by AI systems, concerns about data protection and privacy emerge (Leschke & Salden 2023). It is crucial to establish robust protocols and safeguards to ensure that student information is handled securely and ethically. Additionally, there is a need for transparency and accountability in AI algorithms to mitigate biases and ensure fair and equitable outcomes. Another concern associated with AI in education is the potential reduction of human interaction and social learning experiences (Ayala-Pazmiño 2023). While AI-powered tools can provide personalized instruction and feedback, they may not fully replicate the benefits of face-to-face interactions with teachers and peers. Human interaction fosters social skills, collaboration, and empathy, which are essential for holistic development. Careful consideration should be given to strike a balance between AI-driven personalized learning and the importance of social interaction in educational settings (Liao et al. 2021). In many person-centred degree courses in recent years, the use of VR or AR in particular has shown that human-to-human interaction is irreplaceable in terms of establishing professionalized communication and action strategies. VR and AR training courses are standardized training situations that can provide the basic framework, but they can never do justice to the complexity of human need constellations (Weimann-Sandig 2022). This realization is important because it leads to an admittedly sore point that university lecturers have not had to deal with in recent decades, but which has become increasingly urgent since the digital corona semesters: the question of the self-image and role of teachers and learners in the universities of the future.

3 THE IMPACT OF AI ON THE ROLE OF EDUCATORS AT UNIVERSITIES

Traditionally, educators in Germany, as in many other countries, have held a central role in knowledge dissemination, classroom instruction, and assessment. They have been seen as the primary source of information and guidance for students, responsible for imparting knowledge and facilitating learning. However, the integration of Artificial Intelligence (AI) technologies in education is transforming this traditional role, challenging the existing notions of teaching and learning. The way in which knowledge is acquired is changing more and more. University lecturers are no longer the primary source of knowledge. Students are doing their own research and using generative AI systems such as ChatGPT for this purpose. As a result, many lecturers feel that their knowledge advantage is increasingly being called into question. In terms of a professional self-image or a professional claim in the traditional context, this is entirely understandable. The exclusivity of knowledge and the associated balance of power are central guiding criteria of traditional professional theories (Parsons 1968; Mieg 2018). University lecturers in particular, who conduct a great deal of their own research, define themselves or differentiate themselves from non-experts or recipients of knowledge through this accumulation of expert knowledge. In fact, information and knowledge are two related but distinct concepts: information refers to data or facts presented in a specific form. It is raw, unprocessed elements that can have meaning or significance when interpreted by a person. Knowledge refers to the understanding and interpretation of information. It is the result of processing and internalizing information by an individual. Knowledge encompasses not only the mere presence of information but also the comprehension, connections, and ability to apply and utilize information. It goes beyond mere factual knowledge and includes the capacity to analyze, evaluate, synthesize, and contextualize information. Information becomes knowledge when it is processed, understood, and contextualized by a person (Zoglauer 2021).

In a broader perspective, knowledge can be seen as a complex network of information built through experiences, learning, and reflection. It encompasses not only what we know but also the ability to apply, expand, and update that knowledge. In fact, analyses of teaching concepts and forms of teaching show that, for a variety of reasons (e.g. limited time resources, overcrowded lectures), teachers only provide information in their courses, but often fail to generate knowledge. This requires interactive and modern learning formats, but above all time that allows students to engage discursively and critically with this information. In terms of higher education, however, this is precisely what is essential and what universities actually aspire to. University lecturers should enable students to critically engage with information, examine it, question it or break it down into its individual components so that knowledge can be generated through this process.

The spread of generative AI systems therefore shows us what can be replaced and what we university lecturers actually stand for. Researching information - students can do this very well today with AI such as ChatGPT. Contextualizing, systematizing and ultimately checking this information is something that

students urgently need guidance from university lecturers, as only they have exclusive expert knowledge. University lecturers will therefore not become superfluous through the use of AI, but their role will change and - even more importantly - the teaching formats and examinations will have to change in order for there to be a successful process of knowledge acquisition.

Let's look at the classic example of an introductory course designed as a lecture. Around the world, a hundred or more students crowd into a lecture room, and the subject discipline plays no role here. The students sit close together or even have to stand. A university lecturer monologues far down at the lectern and offers the students orientation options using extensive PowerPoint slides. These contain keywords, i.e. basic information. Using examples, the lecturer tries to link this information and ensure that students acquire critical knowledge. This does not take into account the increasing diversity of students. The classic concept of the lecture in the form of information provision and exemplary in-depth study is always based on the image of a homogeneous student body. Unfortunately, the scope of many universities' diversity and inclusion concepts does not seem to extend to these courses. Otherwise, there should have long been a mandatory orientation towards different learning type variations, cultural characteristics and inclusive learning formats. The importance of the profession of university lecturer arises from the didactic diversity of real knowledge transfer and not the mere provision of information. Promoting critical thinking and problem-solving skills is not only a central goal of higher education, but also essential future skills (Ehlers 2020; Schüller et al. 2019). The integration of AI technologies in the learning environment can offer significant advantages for both educators and students regarding critical thinking, especially if you think beyond ChatGPT and consider other AI-supported systems. Intelligent tutoring systems can provide personalized feedback, adaptive scaffolding and guided practice, promoting deeper engagement and metacognitive skills (D'mello et al. 2013). Virtual reality (VR) and Augmented reality (AR) applications can immerse students in authentic problem-solving scenarios, allowing them to apply their knowledge and analytical abilities in realistic contexts (Sung et al. 2016). Additionally, AI-powered chatbots and virtual assistants can encourage students to ask questions, explore different perspectives, and engage in reflective thinking. The role of university lecturers is changing to encourage students to use these new opportunities or to test them together with them. New tasks and new learning concepts are needed. The critical thinking of university lecturers must change in such a way that they move away from a homogeneous image of students and engage with variations in learning types. This requires time resources but also a desire for a new form of creativity. The monologic processing of Power Point slides has definitely had its day. However, this can only succeed if university lecturers receive support for this. The spread of AI in university teaching is uncharted territory for students and teachers alike, and this should not be ignored. Universities must therefore create new opportunities and testing grounds for teachers so that creative learning formats can be established. In reality, however, many universities offer research sabbaticals but no teaching sabbaticals, and the same applies to reductions in teaching load: these are often granted for research, but only at selected universities for the development of innovative teaching concepts. In the context of higher education in Germany, where a more lecturer-centered approach has traditionally prevailed, the introduction of AI-driven technologies represents a significant shift. Lecturers have historically held the role of content experts, delivering lectures and disseminating information to students (Hüther & Krücken 2018). This approach often featured limited interactivity and individualization. AI technologies challenge this traditional lecturer-centered approach by enabling more student-centered and personalized learning experiences (Weimann-Sandig 2023a). However, this is not a sure-fire success. However, this is not a sure-fire success, but can only succeed if the universities of the future also offer their teaching staff innovative test rooms.

4 INNOVATIVE TEST ROOMS OF THE FUTURE – A POSSIBLE SCENARIO

In order for knowledge transfer to be successful in university teaching, despite or even because of AI, we need to move away from traditional teaching formats such as overcrowded classroom lectures towards new, activating teaching methods. In fact, I repeatedly encounter reservations. There is no substitute for large lectures if you want to qualify the mass of students in the given time. However, my experience with activating teaching methods tells a different story. Information does not have to and should not be provided in lectures, as there are too many rich, AI-supported sources available today - as explained in detail above. Students can obtain this information themselves through self-study, which is becoming increasingly rare at many universities anyway. Here, too, it is the guided procurement and support from lecturers that counts. Lecture times themselves can be used for working on cases or examples in groups, where students have to prove that they can contextualize and analyze information in a peer-to-peer process. Teachers are also continuously available here as coaches and knowledge

mediators. The wealth of digital tools also allows us to move from face-to-face teaching to the digital space or to rely on a combination of both options. But back to the initial idea of enabling and testing spaces for teachers, which absolutely must be increased. In the university development project BediRa (reflexively shaping relationships in the digital space), we are addressing this necessity. One approach to this is the establishment of so-called Scholarships of Teaching and Learning. Scholarships of Teaching and Learning (SoTL) are an approach where educators systematically investigate their own teaching to improve student learning and develop new teaching formats (Almeida 2010; Felten 2013). SoTL involves applying scientific methods to examine the effectiveness of teaching and learning strategies. For this to succeed, universities need to create new structures. First of all, teachers need incentives to break out of their daily routine and embrace something new. Universities can create incentives to encourage participation in SoTL. This may involve recognizing SoTL activities in relation to promotions, tenure processes, or teaching awards. Providing financial support, research grants, or course releases for SoTL projects can also offer incentives. Above all, however, a change in the teaching and learning cultures at universities is needed. It is important to establish SoTL as an integral part of the teaching culture at universities (Felten & Chick 2018). This can be achieved by integrating SoTL into faculty development programs and trainings, creating platforms for the exchange of findings and best practices, and establishing networks and collaboration opportunities among faculty and disciplines. And a third, essential component should not be neglected: scholarships of teaching and learning must not be conceived without the students in mind, but must be designed together with them. Student participation in the sense of students as partners (Cook-Sather, Bovill & Felten 2014) is an important milestone here. After all, it is precisely about creating activating teaching formats that are geared towards the needs of the students. Accordingly, we practice a co-constructivist approach in the BediRa project. In general, constructivism states that knowledge is actively constructed by the learner. In higher education, this means promoting self-directed learning, critical thinking, and discussion. Social-constructivism emphasizes the meaning of interaction and stresses, that cultural contexts play a critical role in learning (Vygotskij 1964). In addition, a sub-strand has derived from this strand, which is called Co-Constructivism. The main difference between constructivism and co-constructivism lies in the emphasis on the social element, i.e. a two-way teaching-learning relationship. While constructivism focuses on individual learning, co-constructivism emphasizes the importance of social interactions and the joint construction of knowledge in social contexts. In the co-constructivist concept, lecturers also become knowledge recipients and thus continue to educate themselves through the input of their students (Weimann-Sandig 2023b). When it comes to integrating AI into higher education by using a co-constructivist approach, Scholarships of Teaching and Learning can be valuable. The interesting thing is that AI can already be helpful in identifying issues for SoTL projects.

CONCLUSION

The transformation of educators' roles through AI in Germany and beyond requires a paradigm shift in the traditional understanding of teaching. Educators need to embrace the potential of AI to enhance their teaching practice and adapt their roles to become facilitators, mentors, and orchestrators of AI-driven learning environments. This shift requires educators to adapt their pedagogical strategies, embrace new roles, and develop skills in effectively utilizing AI tools in their teaching practice. While some educators may initially perceive these changes as a threat to their expertise and authority, AI integration in education should be seen as an opportunity for professional growth and innovation. Educators can focus on higher-order skills such as critical thinking, creativity, and mentorship, fostering deeper engagement and meaningful learning experiences for students. In order for this to succeed, universities must provide their teaching staff with more freedom and scope to try out innovative teaching methods. One possibility here could be so-called Scholarships of Teaching and Learning Through systematic investigations, SoTL enables educators to improve student learning outcomes, develop evidence-based teaching strategies, and contribute to the scholarship of teaching. By examining the integration of AI into higher education through SoTL, universities can harness the potential of AI technologies to enhance student learning experiences.

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